PROGRAM

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_wine
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
# Load dataset
data = load_wine()
# Select 5 numerical features
X = data.data[:, :5]
feature_names = data.feature_names[:5]
print("Features:")
print(feature_names)
# Step 1: Standardization
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
print("\nStandardized Data - First 5 rows:")
print(np.round(X_scaled[:5], 3))
# Step 2: Covariance Matrix
covariance_matrix = np.cov(X_scaled, rowvar=False)
print("\nCovariance Matrix:")
print(np.round(covariance_matrix, 3))
# Step 3: Apply PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
# Step 4: Principal Components
print("\nPrincipal Components:")
print(np.round(pca.components_, 3))
# Step 5: Explained Variance
print("\nExplained Variance:")
print(np.round(pca.explained_variance_, 3))
print("\nExplained Variance Ratio:")
print(np.round(pca.explained_variance_ratio_, 4))
# Step 6: Information retained
retained = sum(pca.explained_variance_ratio_) * 100
print("\nInformation Retained:",
      round(retained, 2), "%")
# Display transformed data
print("\nFirst 5 Transformed Rows:")
print(np.round(X_pca[:5], 3))
# Step 7: Visualization
plt.scatter(X_pca[:, 0], X_pca[:, 1])
```

```
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA - 5 Features Reduced to 2 Components")
plt.show()
```

OUTPUT

```
Features:
['alcohol' 'malic_acid' 'ash'
 'alcalinity_of_ash' 'magnesium']
Covariance Matrix:
[[ 1.006  0.095  0.213 -0.312  0.272]
 [ 0.095  1.006  0.165  0.290 -0.055]
 [ 0.213  0.165  1.006  0.446  0.288]
 [-0.312  0.290  0.446  1.006 -0.084]
 [ 0.272 -0.055  0.288 -0.084  1.006]]
Principal Components:
[[ 0.093  0.429  0.651  0.578  0.222]
 [ 0.658 -0.134  0.184 -0.434  0.572]]
Explained Variance:
[1.640 1.489]
Explained Variance Ratio:
[0.3261 0.2960]
Information Retained: 62.21 %
First 5 Transformed Rows:
[[-0.199  2.719]
 [-2.167  1.168]
 [ 0.614  0.498]
 [ 0.066  2.133]
 [ 1.869  1.040]]
```